son.pu.py - C:/Users/pings/Desktop/son.pu.py (3.14.2)

File Edit Format Run Options Window Help

```
#Z-Test
#2. A tea stall in Chennai sold an average of 400 cups per day with a standard deviation of 45 cups.
#After introducing a new flavour, the owner records sales for a random sample of 36 days
#and finds a mean of 418 cups per day. Test at the 0.05 level of significance whether the new flavour has increased daily sales. (Right-tailed)
#Table value: Z (right-tailed, α = 0.05) = +1.645

import math

mu0 = 400
xbar = 418
sigma = 45
n = 36
alpha = 0.05
table = 1.645

z = (xbar - mu0) / (sigma / math.sqrt(n))

print("Problem 2: Tea stall sales (Z-test, right-tailed)")
print("H0: mu = 400")
print("H1: mu > 400")
print("Level of significance:", alpha)
print("Test statistic: Z =", round(z, 4))
print("Table value: +", table)

if z > table:
    print("Decision: Reject H0")
    print("Conclusion: The new flavour significantly increased daily sales.")
else:
    print("Decision: Fail to reject H0")
    print("Conclusion: The new flavour did not significantly increase daily sales.")
```

Ln: 8 Col: 0

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/son.pu.py ==========
Problem 2: Tea stall sales (Z-test, right-tailed)
H0: mu = 400
H1: mu > 400
Level of significance: 0.05
Test statistic: Z = 2.4
Table value: + 1.645
Decision: Reject H0
Conclusion: The new flavour significantly increased daily sales.
>>>
```

Ln: 13 Col: 0